

Higher-order functions on lists

Higher order functions

- A higher order function is just a function that takes a function as an argument
- We have seen such functions, such as onclick in the turtle module
- Using higher order functions on lists is a useful alternative to loops

The filter function

- `filter(<func>, <iterable>)`
 - `<func>` must return only True or False values
- `filter()` plugs each element of the input `<iterable>` into `<func>` and returns a lazy iterable consisting of only the elements for which `<func>` is true
- Lazy iterables are one time use only

The map function

- `map(<func>, <iterable(s)>)`
 - There must be as many `<iterable(s)>` as `<func>` has inputs
- `map()` plugs all first elements of the `<iterable(s)>` into `<func>` and then all of the second elements and so forth, and returns a lazy iterable consisting of the results of the `<func>` evaluations
- If using multiple `<iterables>`, then the resulting lazy iterable will be the same length as the shortest of the input `<iterables>`

The reduce function

- Not built in - contained in the functools module
- `functools.reduce(<func>, <iterable>)`
 - `<func>` must take two inputs
- Reduce applies `<func>` to the first two elements of `<iterable>`. Then takes the result and applies it to the next element of `<iterable>`, and continues to do so until the iterable is exhausted. The result of the final application of `<func>` is then returned.

Lambda expressions

- Allows you to create “anonymous” functions
- `lambda a: <expression that uses a>`